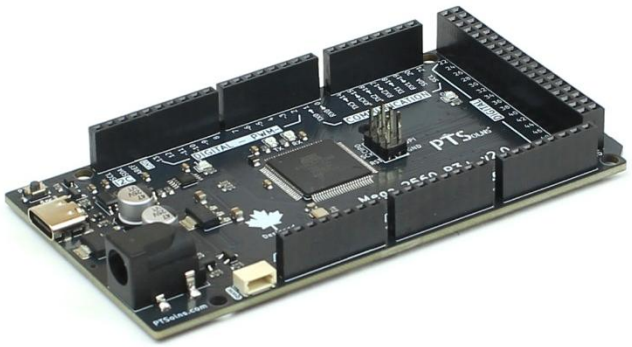
Mega 2560 R3+

1 DESCRIPTION

The PTSolns *Mega 2560 R3+*, based on the ATmega2560, is a popular microcontroller development board with additional unique features. The *Mega 2560 R3+* continues to be compatible with a range of standard shields, offering the standard Arduino Mega 2560 pin numbering and layout. A standout feature is the onboard QWIIC® connector, which provides a shifted 3.3 V I2C bus, making it convenient to connect many common sensors and modules that operate on 3.3 V logic. There is no need for extra external components; all the logic level shifting between 5 V and 3.3 V on the I2C bus is done automatically onboard.



The *Mega 2560 R3+* replaces the previous bulky port with a modern USB-C port, while the standard power barrel jack remains onboard. The board has a streamlined layout of components, including a clean grouping of the onboard LEDs (ON, 13, TX, RX). Furthermore, the *Mega 2560 R3+* allows for a larger current draw on the 3.3 V pin with a rated maximum of 0.8A. This lets the user manage their project's power budget more easily by allowing a larger current draw.

The *Mega 2560 R3+* comes with a pre-installed out-of-the-box ready set of tests. This gets simple projects started quickly and allows for testing the proper function of the board. A large active online community supports this microcontroller and its programming using *PTSolns IDE* or the Arduino IDE software.

This *Plus* version offers all of the features users come to learn and expect and adds additional ones that make this board stand out.

All *Mega 2560 R3+* boards are individually inspected and marked with a quality control sticker (either on the PCB, or on the packaging).

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2 DOCUMENT REVISION HISTORY

Current document revision is Rev 0.

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3 PRODUCT FEATURES

This section highlights notable features of the *Mega 2560 R3+*.

3.1 I/O Connectors & Input Power

The *Mega 2560 R3+* has three input/output (I/O) connectors as follows:

- USB-C
- Power barrel jack (2.1 mm x 5.5 mm)
- QWIIC®

These connectors are shown in Figure 1. Note that the QWIIC connector adheres to the standardization as outlined by SparkFun Electronics: https://www.sparkfun.com/qwiic

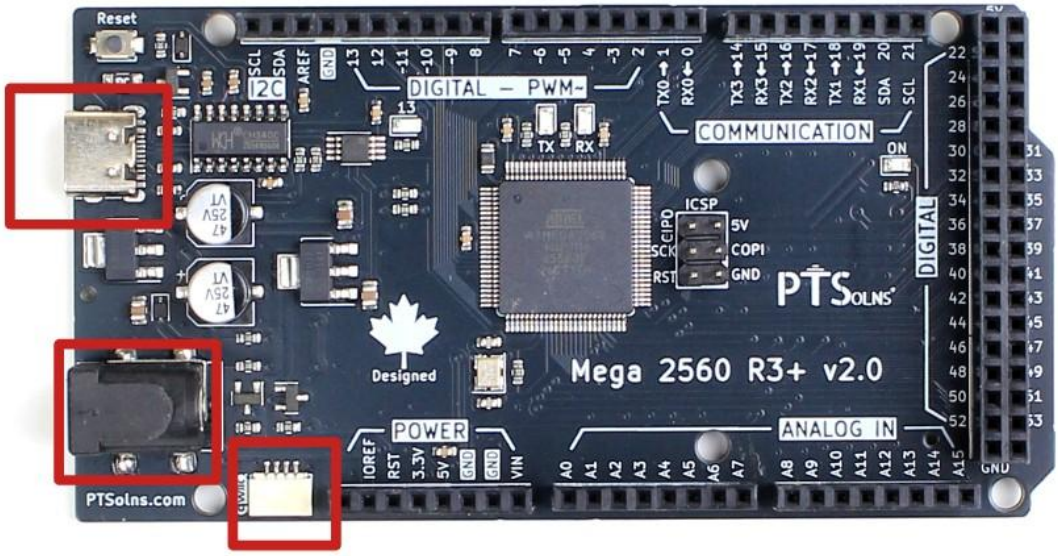


Figure 1: The three I/O connectors on the Mega 2560 R3+.

The *Mega 2560 R3+* can be powered in multiple ways, with the most common two methods being the USB-C port and the power barrel jack. If the user connects both at the same time, the board automatically selects the voltage from either source depending on the voltage level of the power jack. The user can also power the board using the "VIN" pin on the female POWER header with the same voltage input ratings as for the power barrel jack (i.e., Vin = 7-12 V max).

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3.1.1 Note on USB-C Cables

From a data transfer perspective USB-C cables can be categorized into two types:

1. Data transfer capable
2. Not data transfer capable

Only the first type of USB-C cable can be used to program not just the *Mega 2560 R3+*, but indeed any microcontroller. This type of cable provides power to the board, as well as facilitates data transfer between the computer and the board. **Using this type of cable is essential in programming a microcontroller**.

The second type of USB-C cable does not facilitate the transfer of data but can merely be used to power the board. Therefore, this type of USB-C cable cannot be used to program a microcontroller.

How to tell if a USB-C Cable is Data Transfer Capable?

One can easily and quickly check if a particular USB-C cable is data transfer capable by simply trying to program the *Mega 2560 R3+*. If the USB-C cable is not data transfer capable, then upon plugging it into the computer with the other end into the *Mega 2560 R3+* no port will appear.

3.2 I2C Bus

The *Mega 2560 R3+* offers an I2C bus on the SDA and SCL lines (Pins 20 and 21, as well as the dedicated I2C header block), inherently at the ATmega2560 5 V logic level.

The board also features an onboard Logic Level Shifter (LLS) that shifts the 5 V I2C bus down to 3.3 V. This shifted 3.3 V I2C bus is made available via the QWIIC connector located next to the power barrel jack.

The default connection for the 3.3 V QWIIC connector is enabled. However, it is possible to disconnect these pins by cutting the jumper pads on the back of the board labeled **"3.3V SDA"** and **"3.3V SCL"**. Furthermore, there are additional pull-up resistors on the 5 V I2C bus. These can also be disconnected by cutting the jumper pads labeled **"10k Pull-up 5V SCL"** and **"10k Pull-up 5V SDA"** on the back of the PCB.

The user can connect multiple peripherals to the *Mega 2560 R3+* via I2C simultaneously. It is possible to connect to the 5 V I2C, shifted 3.3 V I2C as well as the QWIIC connector all at once as long as the following requirements are met:

- The total number of I2C connected peripherals does not exceed 126 units.
- The power budget of the project does not exceed any current rating (See Section 5).
- Each device address is unique (alternatively an I2C MUX (https://ptsolns.com/products/i2connect-mux) can be used to connect peripherals with the same address on the same I2C bus).
- There is not too much capacitance introduced by long connecting wires. The rise-time of the I2C signals as determined by the corresponding time constant must be within acceptable range (See discussion below).

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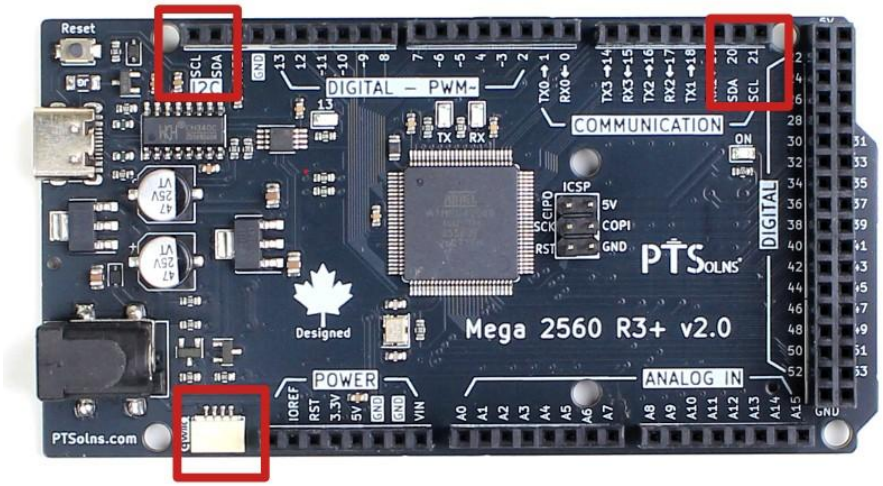


Figure 1: I2C pins at 5 V and 3.3 V, as well as the QWIIC connector, on the Mega 2560 R3+.

The default connection for the 3.3 V SDA and SCL pins on the Mega 2560 R3+ is such that it is enabled. This means the Mega 2560 R3+ out-of-the-box will have a functioning 3.3 V I2C bus available exclusively via the QWIIC connector, as seen near the bottom-left in Figure 1. However, it is possible to disconnect these pins by cutting the jumper pads labelled “3.3V SDA” and “3.3V SCL” as shown in Figure 2. When these pins are cut, the QWIIC connector's I2C pins are floating.

Furthermore, there are additional pull-up resistors on the 5 V I2C bus each of 10 kΩ. These can also be disconnected by cutting the jumper pads labelled “10k Pull-up 5V SCL” and “10k Pull-up 5V SDA” as shown in Figure 2. A full schematic of the Logic Level Shifter (LLS) showing the 5 V and the 3.3 V sides of the I2C bus, as well as the four jumper pads, can be seen in Figure 3.

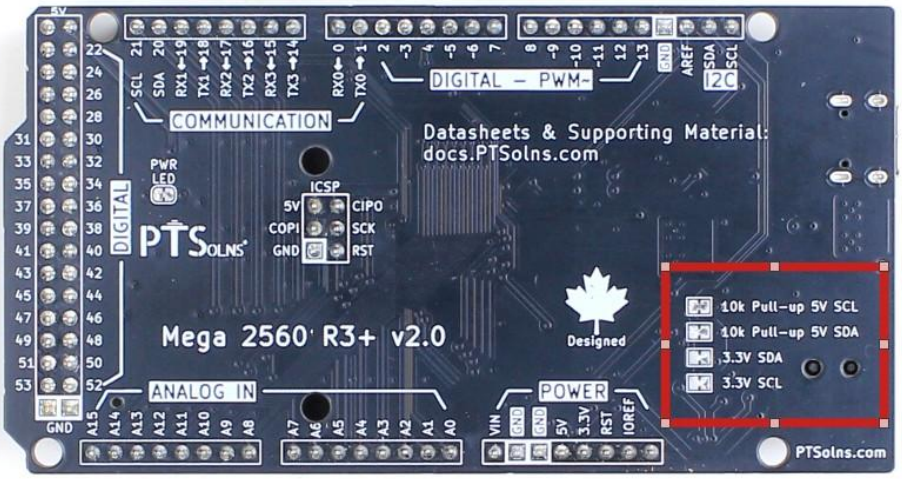


Figure 2: Jumper pads on the back of the Mega 2560 R3+ (all closed by default).

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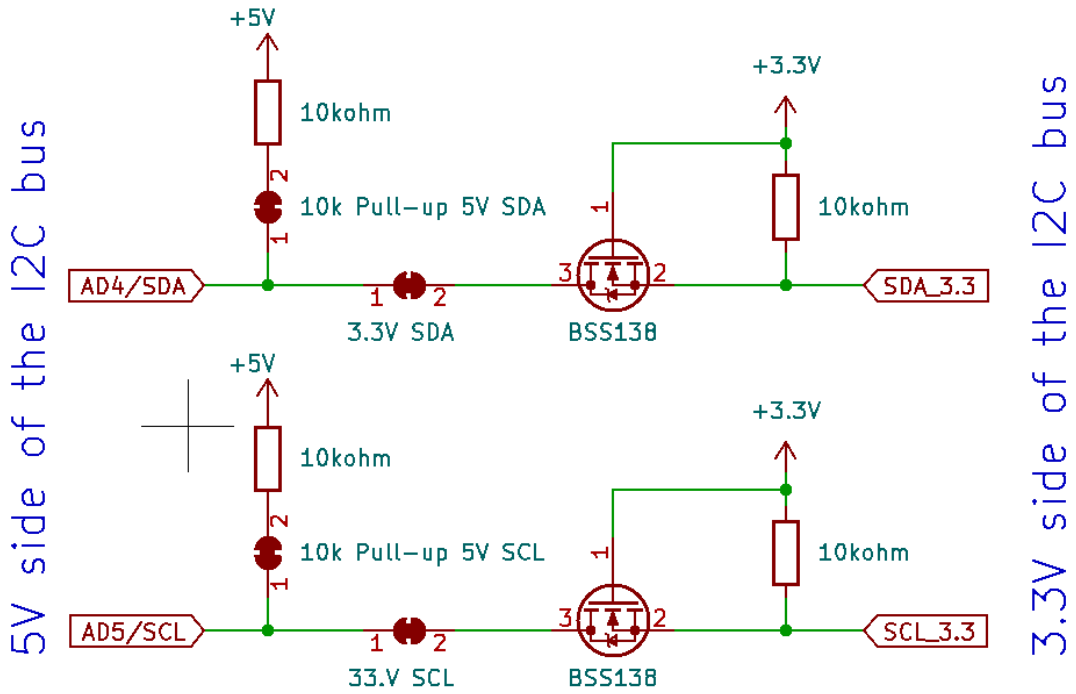


Figure 3: Logic Level Shifter (LLS) schematic on the Mega 2560 R3+ showing the 5 V and the 3.3 V sides of the I2C bus.

The user is made aware that if too many I2C connected peripherals are used at once, or the connecting wires are long (around 1.5m or longer), the I2C communication may start to break down. This is not a deficiency of the *Mega250 R3+* but rather a characteristic of I2C communication itself. When the SDA signal rise-time starts to take too long and becomes sluggish (because the time constant is too large, which is a function of resistance and capacitance on the I2C bus), then data may become corrupt. Often this will result in the microcontroller freezing and a restart may be required. This scenario is easily diagnosed with an oscilloscope. The user can try the following simple changes to improve the I2C bus quality:

- Reduce the communication wire on the SDA and SCL lines.
- Add or decrease pull-up resistors on the SDA and SCL lines. Note that many peripherals have their own pull-ups onboard, and when connecting many of such modules on the I2C bus that the total resistance acts as if in parallel, thus decreasing the overall resistance. The user may have to decrease the resistance or disable entirely. Alternatively, there may not be enough resistance and so larger resistance may need to be added.
- Reduce the clock speed of the SCL line. The slower the communication speed, the less sluggish or slow rise-time of the SDA line impacts the communication. Typically, the default SCL clock speed is 100 kHz. Although there is no theoretical lower limit for I2C, some peripherals have issues working below 10 kHz.

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3.3 ICSP Breakout

The *Mega 2560 R3+* has an In-Circuit Serial Programming (ICSP) breakout section via a 2X3 Pin male header, as shown in Figure 4. The pin breakout of the ICSP is shown in the schematic in Figure 5, with the following definitions:

- CIPO: Controller In, Peripheral Out (formerly called MISO)
- SCK: Serial Clock
- COPI: Controller Out, Peripheral In (formerly called MOSI)
- Reset: Reset

For orientation of the pins, the ground “GND” pin is outlined with a white printing, best seen from the back of the board.

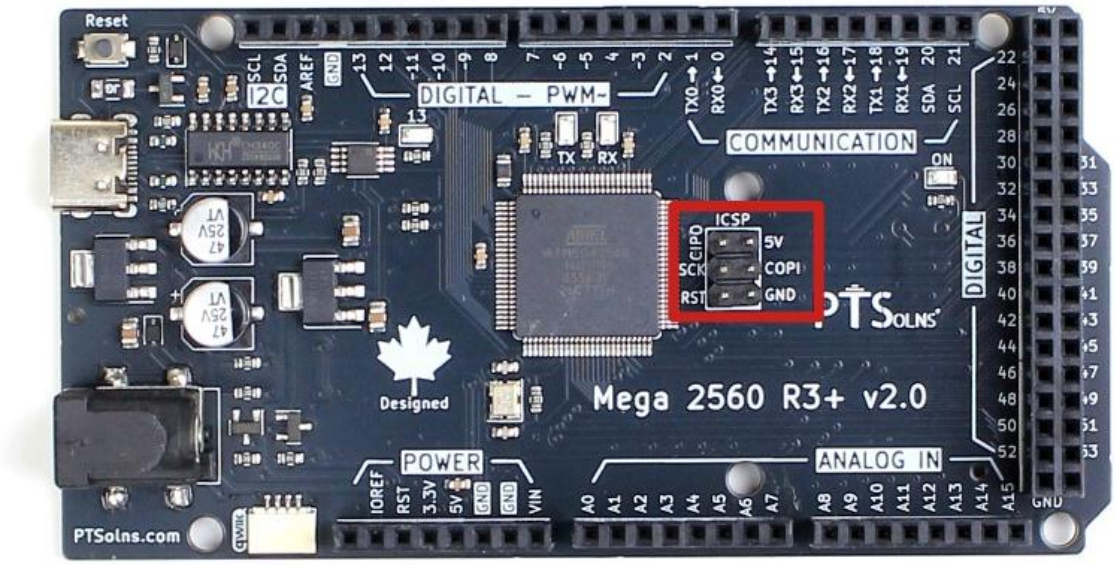


Figure 4: Breakout pins of the Mega 2560 R3+.

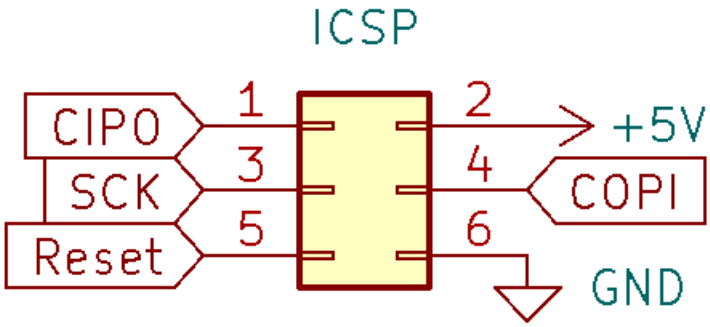


Figure 5: ICSP breakout pins.

3.4 Compatibility with Shields (Stackability)

The *Mega 2560 R3+* is fully stackable and thus compatible with a range of common shields. Therefore, all the regular breakout pins both be the female headers as well as the male header are available, as shown in Figure 4.

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3.5 Component Arrangement & Pinout Diagram

The *Mega 2560 R3+* has a unique component arrangement. As an example, the four onboard LEDs (Power LED, Pin 13 controlled LED, TX and RX communication LEDs). Some notable component arrangements, including the LEDs, Reset button and the ATmega2560, are shown in Figure 6.

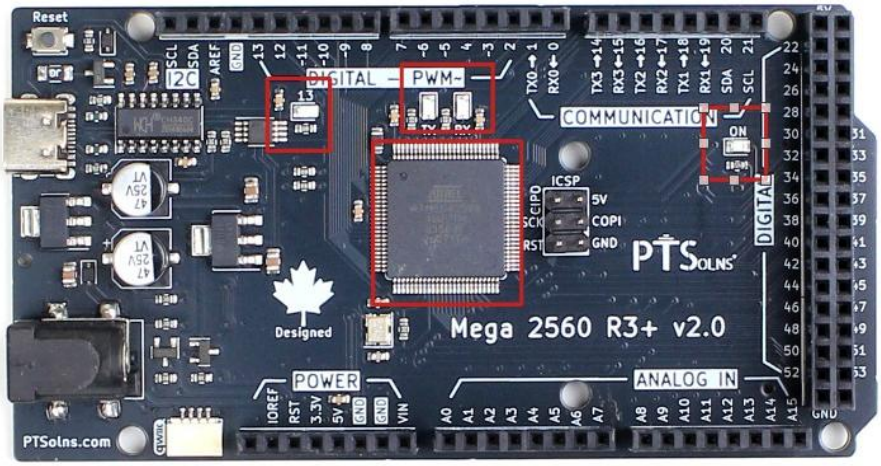


Figure 6: Notable component arrangements of the Mega 2560 R3+.

The *Mega 2560 R3+* has 54 digital Input/Output (I/O) pins, referred to as "DIO". This includes 14 digital Input/Output (I/O) pins grouped on the top header, of which 12 are PWM capable, plus an additional 32 digital I/O pins on the right-side header. (Across the entire board, 15 pins are inherently hardware PWM capable). There are 16 analog input pins, 4 hardware serial ports (UARTs), and various power-related pins. All of the pins are shown in the pinout diagram in Figure 7.

Note the following:

- DIO 13 also controls the onboard LED labelled "13" as shown.
- The Reset pin is connected to the push button as well as the pinout on the ICSP.

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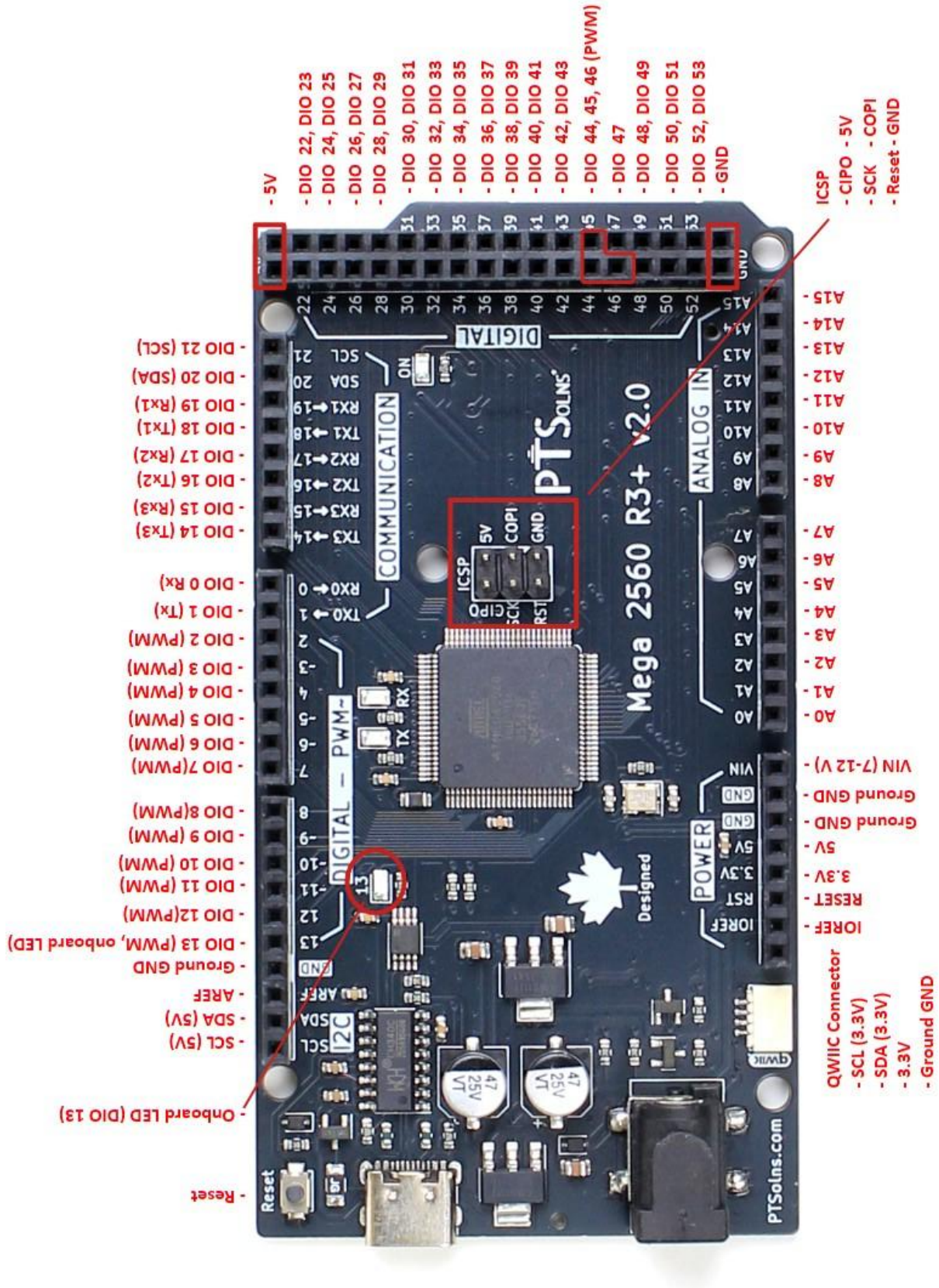


Figure 7: Pinout diagram for the Mega 2560 R3+.

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3.6 Mark of Authenticity

Authentic PTSolns PCBs have a black solder mask color and are marked with the “PTSolns” logo in white silkscreen printing. The “Canadian Designed” symbol, consisting of the Canadian Maple Leaf with the word “Designed” underneath, can also be found on the PCB in white silkscreen printing. The “PTSolns” trademark and the “Canadian Designed” symbols are shown in Figure 8.

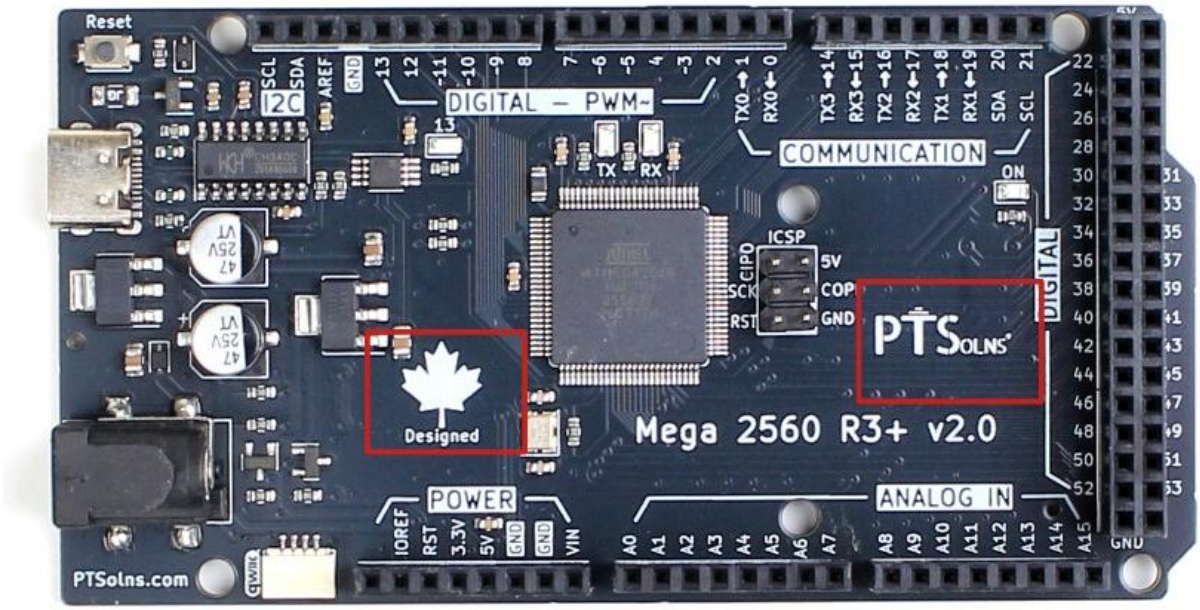


Figure 8: The "Canadian Designed" symbol found on authentic PTSolns PCBs.

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4 PHYSICAL PROPERTIES

The physical properties of the *Mega 2560 R3+* are outlined in Table 1.

Table 1: Physical Properties.

	Quantity	Value	Reference
PCB	Length (longest)	102.55 mm	Figure 9
	Width (longest)	53.3 mm	Figure 9
	Thickness	1.6 mm	--
	Weight (with IC)	33 g	--
	Color	Black	--
	Silkscreen	White	--
Material	Lead free HASL-RoHS surface finish		--
	FR-4 base		--
Mounting Holes	6x each with 3.2mm diameter		Figure 10

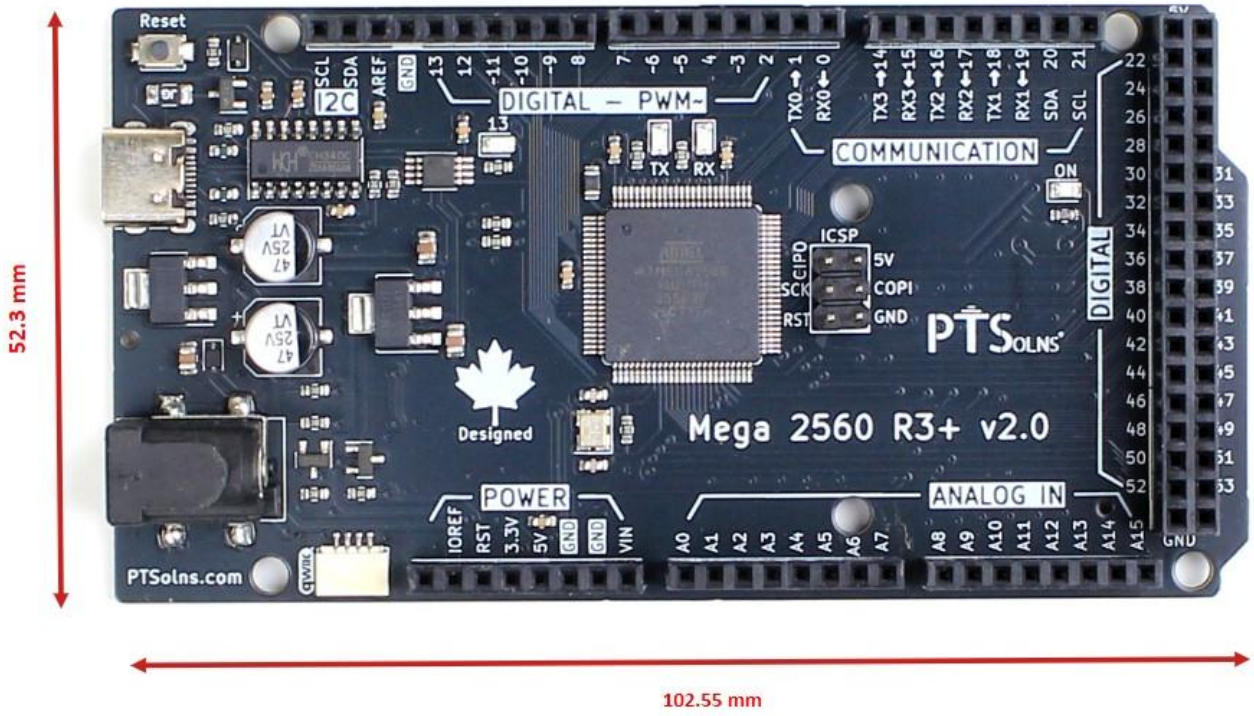


Figure 9: Dimensions of the Mega 2560 R3+.

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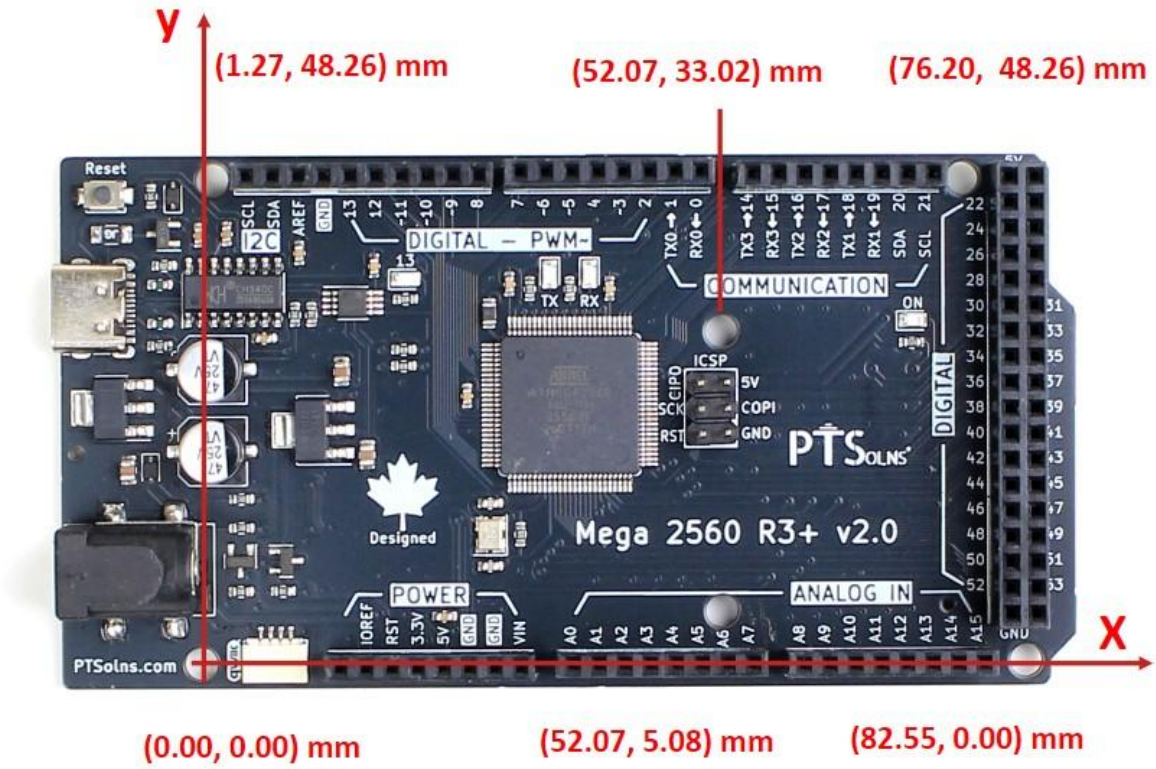


Figure 10: Positions of mounting holes. Coordinate grid zeroed on the bottom-left mounting hole.

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5 ELECTRICAL PROPERTIES

The electrical ratings for the *Mega 2560 R3+* are outlined in Table 2.

Table 2: Electrical ratings for the *Mega 2560 R3+*.

Type of trace	**Rating**
Input voltage on USB-C	5 V
Input voltage on power barrel jack	7-12 V
Operating current draw on any single GPIO	20 mA
Max current on any single GPIO	40 mA **(Do not operate at this level for extended periods)**
Max combined current on all GPIO	200 mA **(IF OPERATING in "Stable" CONDITIONS. See Section 5.1 for important details)**
Max current draw on 3.3 V pin	800 mA **(IF OPERATING in "Stable" CONDITIONS. See Section 5.1 for important details)**
Max current draw on 5 V pin	800 mA **(IF OPERATING in "Stable" CONDITIONS. See Section 5.1 for important details)**
Max combined current draw on all GPIO and power pins (Total External Current Draw (TECD))	800 mA **(IF OPERATING in "Stable" CONDITIONS. See Section 5.1 for important details)**

The current ratings for the two power pins, the 3.3 V and 5 V pins, are discussed in more detail in Section 5.1. The voltage ratings are discussed in Section 5.2.

5.1 Current Rating

The onboard components on the *Mega 2560 R3+* consume approximately 20 to 40 mA, depending on the LEDs and the Atmega2560 microcontroller demand. External current draws can be made by employing any of the General-Purpose Input/Output (GPIO) pins, such as the digital or analog pins, as well as the 3.3 V and 5 V power pins. The 3.3 V and 5 V power pins can be used to power external sensors and modules. The combined current draw of any GPIO and power pins used is the Total External Current Draw (TECD).

The TECD is delivered by an onboard 5 V voltage regulator. Depending on the input voltage (Vin) supplied to the regulator, as well as the TECD, the *Mega 2560 R3+* may operate in the "Stable" region or the "Unstable" region. This is shown in Table 3. In this context, "Stable" is defined such that the 5 V line remains constant, with a small ripple, within acceptable tolerances. "Unstable" is defined such that the 5 V line starts to drop below unacceptable tolerances, collapses, or rises above 5 V plus an acceptable tolerance. The user should only operate the *Mega 2560 R3+* in the "Stable" region. In extreme unstable regions (e.g. Vin = 12 V, TECD = 800 mA), the voltage regulator may allow the input voltage through to the 5 V line. This can cause damage to components downstream, and the user must avoid such extreme conditions at all times.

As an example, at an input voltage of Vin = 7 V, the TECD can reach the maximum of 800 mA. With increasing input voltages, the TECD in which the *Mega 2560 R3+* operates in the "Stable" regions starts to reduce.

Therefore, the user should take care that all current draws on external pins (GPIO, 3.3 V and 5 V power pins) remains in the "Stable" region for a given input voltage Vin, as outlined in Table 3.

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Table 3: TECD Operating Conditions.

Total External Current Draw (TECD)

(Not including power draw of onboard components)

Vin	0.0 A	0.1 A	0.2 A	0.3 A	0.4 A	0.5 A	0.6 A	0.7 A	0.8 A
7.0 V	Stable	Stable	Stable	Stable	Stable	Stable	Stable	Stable	Stable
7.5 V	Stable	Stable	Stable	Stable	Stable	Stable	Stable	Unstable	DNO
8.0 V	Stable	Stable	Stable	Stable	Stable	Stable	Unstable	DNO	DNO
8.5 V	Stable	Stable	Stable	Stable	Stable	Unstable	DNO	DNO	DNO
9.0 V	Stable	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO
9.5 V	Stable	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO
10.0 V	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO	DNO
10.5 V	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO	DNO
11.0 V	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO	DNO
11.5 V	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO	DNO
12.0 V	Stable	Stable	Stable	Unstable	DNO	DNO	DNO	DNO	DNO

Stable	Voltage on 5 V line remains stable.
Unstable	Voltage on 5 V line collables and becomes unstable.
DNO	Do No Operate!

NOTE 1: Drawing full allowed current and temperature of 5 V voltage regulator

If the TECD is high the voltage regulator will get hot. This is unavoidable and depends greatly on the input voltage. The component's temperature depends on how much power (in Watts) it has to dissipate. This power is a result of the voltage difference between the Vin and Vout of the voltage regulator, times the current being draw. This can be expressed in the following formula:

$$Power_{dissipate\ on\ volt.reg} = (V_{in} - V_{out}) * I$$

The input voltage (discussed more in Section 5.2) ranged from 7 V to 12 V. The most power entering in the voltage regulator needing to be dissipated in terms of heat is when Vin = 12 V and the TECD is the maximum of 800 mA. In that case, the power to be dissipated becomes (12 V – 5 V)*800 mA = 5.6 W. This is a lot of power for such a small SMD component. The component will heat up! The 5 V voltage regulator will automatically shut off when the internal component temperature reaches 145 degrees C. The component restarts when the temperature is below a threshold.

The *Mega 2560 R3+* was developed with the temperature and current ratings in mind. The traces within the PCB that carry the current have been made very wide to reduce the temperature. This is particularly the case for the trace to the 5 V power pin. Furthermore, thermal vias were added below both the 5 V and 3.3 V voltage regulator. These vias take the surplus heat from under the component and bring it to the other side of the PCB where there is more surface area to dissipate the heat.

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5.2 Voltage Rating

The 5 V voltage regulator determines the maximum and minimum acceptable voltage input a user can supply. The maximum is set to 12 V. However, the regulator can accept short momentary voltage spikes up to 14 V caused by the external power source. The user should be careful to never exceed the 12 V rating as otherwise damage to the components can result. If the input voltage is too high the regulator can malfunction, allowing high input voltage on the 5 V line, causing other components downstream on the 5 V line to be damaged, including the ATmega328P. Therefore, if wanting to supply the *Mega 2560 R3+* with a 12 V source, particularly from a battery or an unstable buck/boost converter, ensure that the voltage is not above the 12 V rating.

The minimum input voltage is specified as 7 V. With 7 V the *Mega 2560 R3+* can reliably be operated. That being said, the voltage can likely be a little bit lower. It depends on how much current is being drawn through the 5 V voltage regulator, as well as the fuse settings on the ATmega328P. The higher the current draw, the larger the dropout voltage is. At 800 mA current draw, the regulator drops ~1.45 V. An input voltage of 6 V might work well for very low external current draws. The user is encouraged to experiment if in their project setup a lower input voltage is acceptable.

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6 PROGRAMMING

This section explains the first-time setup of the *Mega 2560 R3+*, including driver setup, USB-C cable power and data requirements, and programming (uploading a sketch onto the board).

6.1 USB-C Cable, Data Transfer Capable

From a data transfer perspective, USB-C cables can be categorized into two types:

1. Data transfer capable
2. Not data transfer capable

Only the first type of USB-C cable can be used to program not just the *Mega 2560 R3+*, but indeed any microcontroller. This type of cable provides power to the board, as well as facilitates data transfer between the computer and the board. **Using this type of cable is essential in programming a microcontroller**.

The second type of USB-C cable does not facilitate the transfer of data but can merely be used to power the board. Therefore, this type of USB-C cable cannot be used to program a microcontroller.

How to tell if a USB-C Cable is Data Transfer Capable?

One can easily and quickly check if a particular USB-C cable is data transfer capable by simply trying to program the *Mega 2560 R3+*. If the USB-C cable is not data transfer capable, then upon plugging it into the computer with the other end into the *Mega 2560 R3+* no port will appear.

To read more about data transfer capable USB-C cables, the reader is referred to our Tinker Thoughts Blog TTB#10, and related video:

https://ptsolns.com/blogs/tinker-thoughts/ttb10-why-your-usb-c-cable-wont-program-your-microcontroller

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6.2 *PTSolns IDE* to Program Development Boards

Programming the *Mega 2560 R3+*, in fact most common microcontroller development boards, is easy and intuitive with the programming software *PTSolns IDE*. It is free to download and use. Open-source, community driven, and hosted on GitHub. There are **no sign-ups**, **no sign-in**, and **no subscriptions**. The desktop app is available for Windows, Apple, and Linux operating systems (OS). *PTSolns IDE* can be downloaded from the following link:

https://PTSolns.com/IDE

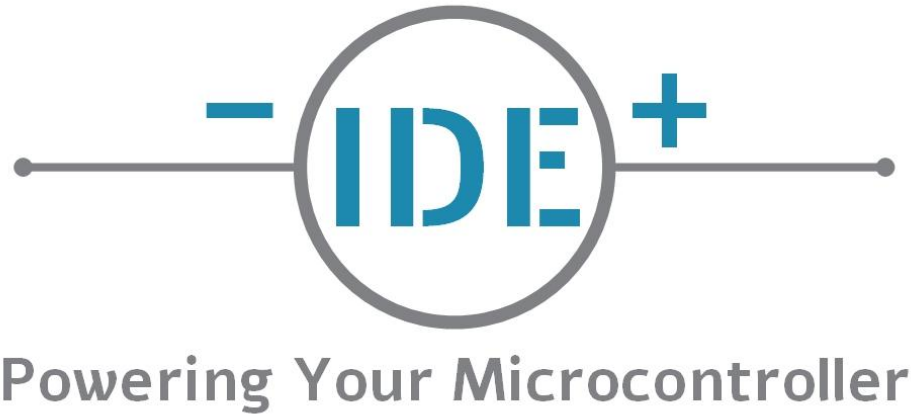


Figure 11: PTSolns IDE - Powering Your Microcontroller

For help with anything related to installation or first time launching the software, please consult this page:

https://PTSolns.com/IDE-Help

6.2.1 CH340 Driver

The *Mega 2560 R3+* uses the common CH340 IC to facilitate communication between what is plugged into the USB-C port (e.g. user's laptop) and the microcontroller (ATmega2560). This IC is required when programming the *Mega 2560 R3+.* The driver for the CH340 typically must be installed the first time it is needed in any project. Many boards and modules make use of the CH340 so chances are that the driver is already installed. However, if the driver is not yet installed, the user must first install it to program the *Mega 2560 R3+.*

It is fast and easy to install the CH340 driver directly from within *PTSolns IDE*. Navigate to the top bar menu and select "Tools" and then "Install CH340 Driver" as shown in Figure 12. This will trigger the CH340 driver installation that only takes a minute or two.

For help relating to installing and checking the CH340 on Windows or macOS, please consult this page:

https://PTSolns.com/CH340

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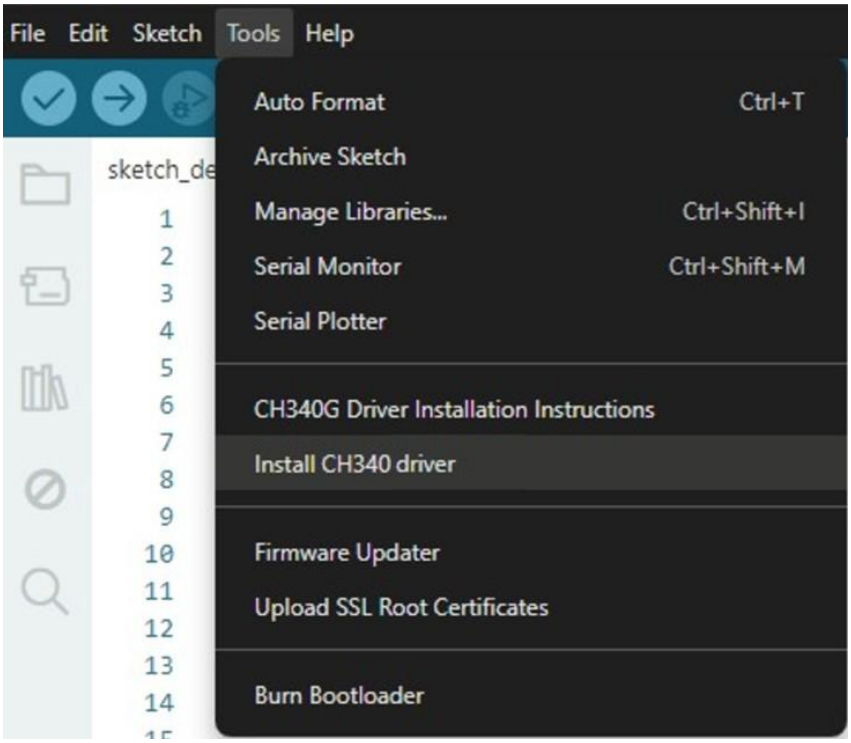


Figure 12: Installing CH340 Driver Directly from within PTSolns IDE.

6.2.2 Board Selection in *PTSolns IDE*

When first opening *PTSolns IDE* the *Mega 2560 R3+* board must be selected. To do so, press the down arrow on the Boards and Ports window, as shown in Figure 13. Upon pressing the little pencil icon, or equivalently the text "Select other board and port…", a new window appears. In the search, type in "mega" and the board selection comes up, as shown in Figure 14.

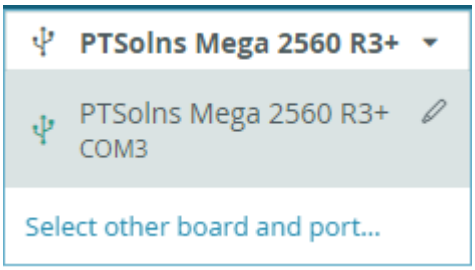


Figure 13: Boards and Ports Selection in PTSolns IDE.

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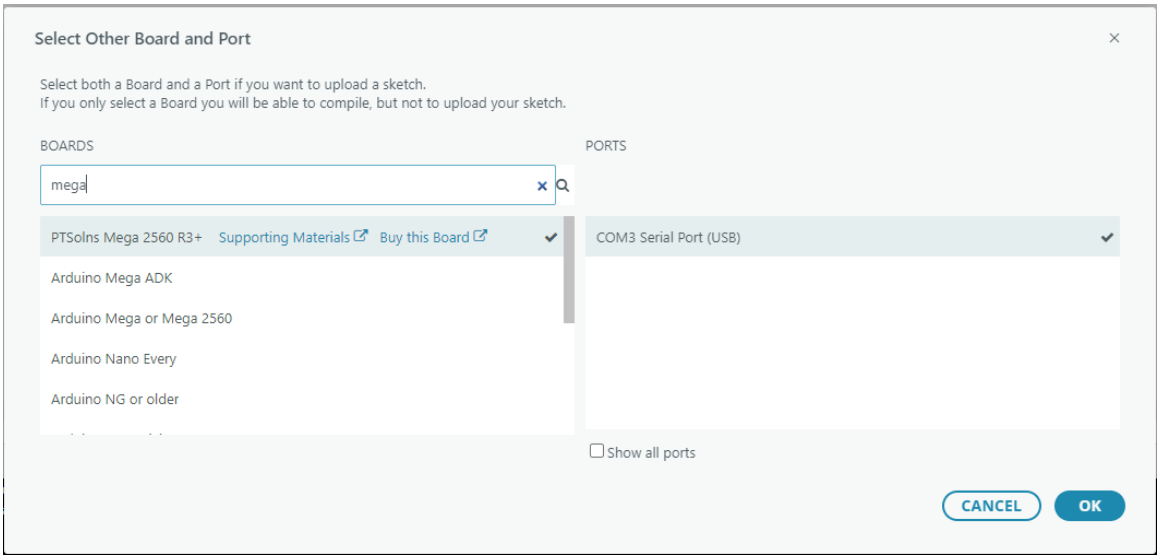


Figure 14: Select the Mega 2560 R3+ Board in PTSolns IDE.

Note that in the image of Figure 14 on the left side is the selection for the BOARDS (*Mega 2560 R3+* in this case), and on the right side the selection for the PORTS. The port selection and details relating to it are explained in Section 6.2.3 .

6.2.3 Port Selection in *PTSolns IDE*

To be able to see one or more ports in the PORTS selection section in Figure 14 (on the right), the following two conditions must be true:

1. The *Mega 2560 R3+* is plugged into the computer with a USB-C cable that is capable to transfer data. Some USB-C cables only provide power, but is not able to transfer data. For more information on this the user is referred to Section 6.1 .
2. The CH340 driver must be installed. See Section 6.2.1 .

If the above two conditions are satisfied, the user will be able to see a port selection on the right side of the image in Figure 14. It is possible to see more than one port to choose from, in that case the user can simply unplug and re-plug the board and see which port number momentarily went offline and came back.

6.2.4 Processor Selection in *PTSolns IDE*

The *Mega 2560 R3+* can be used with the processor “ATmega2560 (Mega 2560)” selection.

6.2.5 Running First Example Sketch *PTSolns IDE*

A common and popular sketch to run on a new development board is the classic “Blink” example. This sketch is useful as the coding is simple, yet the successful upload and running demonstrates critical working parts of the software as well as the board itself. Running Blink demonstrates that:

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- *PTSolns IDE* is working properly,
- The CH340 driver is installed and able to recognize the board, and
- The hardware onboard the development board is working (USB to microcontroller, power circuitry)

To select the Blink sketch is simple from within *PTSolns IDE* as it is installed by default. Click on “File” in the top bar menu, then “Examples\01.Basics\Blink”, as shown in Figure 15.

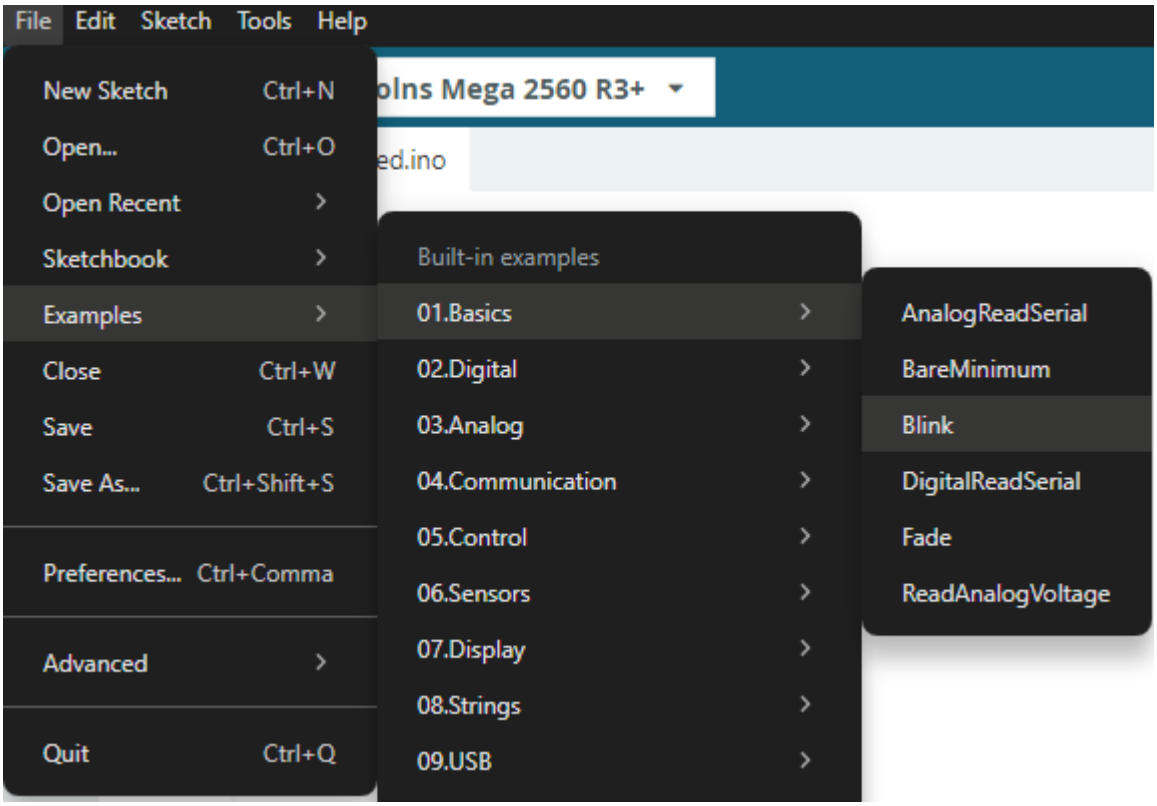


Figure 15: Selecting the "Blink" example from within PTSolns IDE.

6.3 Arduino IDE to Program Development Boards

Arduino IDE is an application to program microcontroller development boards. The application can be downloaded from the following link:

https://www.arduino.cc/en/software

6.3.1 CH340 Driver

The *Mega 2560 R3+* uses the common CH340 IC to facilitate communication between what is plugged into the USB-C port (e.g. user’s laptop) and the microcontroller (ATmega2560). This IC is required when programming the *Mega 2560 R3+.* The driver for the CH340 typically must be installed the first time it is needed in any project. Many boards and modules make use of the CH340 so chances are that the driver is already installed. However, if the driver is not yet installed, the user must first install it to program the *Mega 2560 R3+.*

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Arduino IDE does not inherently support the CH340 driver, therefore it must be installed externally as a standalone program. For installation instructions the user is referred to our installation video:

https://www.youtube.com/watch?v=UUQ84VKg3oM

Additionally, our version of the CH340 installer is available as a standalone tool. It can be downloaded from the following URL (under header “Downloading the CH340 Driver Manually”):

https://PTSolns.com/CH340

6.3.2 Board Selection in *Arduino IDE*

When first opening Arduino IDE the *Mega 2560 R3+* board must be selected. Arduino IDE does not inherently have the exact board selection option, however the board “Arduino Mega or Mega 2560” works equally well, as shown in Figure 16 and Figure 17.

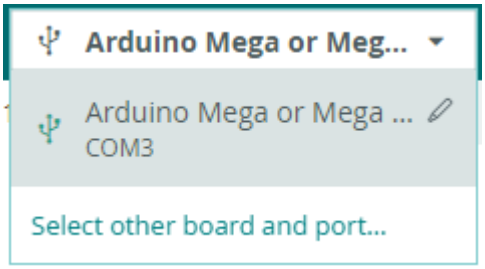


Figure 16: Select other board and port...

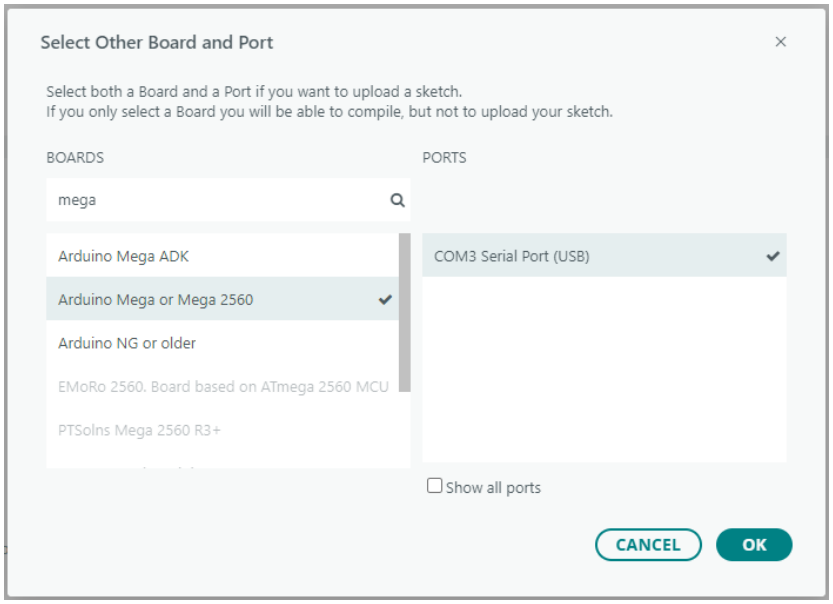


Figure 17: "Arduino Mega or Mega 2560" board selection for the Mega 2560 R3+.

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6.3.3 Port Selection in *Arduino IDE*

To be able to see one or more ports in the PORTS selection section in Figure 17 (on the right), the following two conditions must be true:

3. The *Mega 2560 R3+* is plugged into the computer with a USB-C cable that is capable to transfer data. Some USB-C cables only provide power, but is not able to transfer data. For more information on this the user is referred to Section 6.1 .
4. The CH340 driver must be installed. See Section 6.2.1 .

If the above two conditions are satisfied, the user will be able to see a port selection on the right side of the image in Figure 17. It is possible to see more than one port to choose from, in that case the user can simply unplug and re-plug the board and see which port number momentarily went offline and came back.

6.3.4 Processor Selection in *Arduino IDE*

The *Mega 2560 R3+* can be used with the processor "ATmega2560 (Mega 2560)".

6.3.5 Running First Example Sketch *Arduino IDE*

Running the first example is straightforward and similar to *PTSolns IDE*. Therefore, the user is referred to Section 6.2.5 .

6.4 Out-of-the-Box Ready Examples

The *Mega 2560 R3+* comes pre-installed with a sketch that allows the user to perform several tests without any initial software uploading or programming. These pre-programmed tests can be used to check the working order of the *Mega 2560 R3+,* or to get started quickly making some simple examples. These out-of-the-box ready tests include:

1) Onboard (and Pin 13) LED blinking in unique pattern.
2) Reset makes onboard LED (and Pin 13) blink in fast pattern of four for one cycle.
3) I2C scanner searches for connected devices (5 V or 3.3 V I2C bus, or QWIIC connector) every 5 seconds and prints the results to Serial monitor on baud rate 9600.
4) Pin 9 is on a fading in and out cycle that can drive an external LED accordingly.
5) Analog read on Pin A0 and displayed to Serial monitor on baud rate 9600.

Each of these tests is explained in further detail below. If the user wants to restore the pre-installed testing sketch, it can be downloaded from the PTSolns documentation repository sub-domain:

https://docs.ptsolns.com/Products/PTS-00215_Mega_2560_R3_Plus/Sketches/Mega_2560_R3plus_Getting_Started.ino

6.4.1 Test 1: Onboard LED (Pin 13) Blink Unique Pattern

With the *Mega 2560 R3+* powered, the programmable onboard LED (marked "13") blinks in a regular unique pattern. The LED will illuminate for 100 mS and turn OFF for 200 mS. If the reset is pressed (see Test 2) the pattern changes momentarily before returning to the same pattern.

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The programmable onboard LED (marked “13”) is also connected to Pin 13, available on one of the pins of the female header (marked “13”). As a further test, the positive terminal of a standard LED can be put onto the female header Pin 13, with the negative terminal of the LED attached to a resistor (in the range of ~200 Ω to 1000 Ω, give or take). The other free end of the resistor can be plugged onto one of the ground (marked “GND”) pins in a female header. The LED and resistor can either be soldered together, or a breadboard can be used to make the electrical connection.

6.4.2 Test 2: Onboard LED (Pin 13) Blink Reset Pattern

The reset button triggers a momentarily different pattern consisting of four rapid blinks of the onboard LED (marked “13”) of 50 mS ON and 50 mS OFF. This tests that the *Mega 2560 R3+* is restarting properly.

Upon reset the *Mega 2560 R3+* output several messages to the Serial monitor on baud rate 9600. To see these messages, load the Arduino IDE software and plug in the *Mega 2560 R3+* (ensure that the CH340 driver is installed). Turn on the Serial monitor (select baud rate 9600) and read the output window.

6.4.3 Test 3: I2C Scanner

Every five seconds the I2C bus (20/SDA and 21/SCL) is scanned for any connected peripherals. The I2C bus is available on the female header pins (See Section 3.5 for the pinout diagram). The results of the scan are printed to the Serial monitor on baud rate 9600. To see these results, load the Arduino IDE software and plug in the *Mega 2560 R3+* (ensure that the CH340 driver is installed). Turn on the Serial monitor (select baud rate 9600) and read the output window.
The user can connect several I2C peripherals at once. All the device addresses will be displayed in the Serial monitor.

6.4.4 Test 4: Pin 9 Fade

Pin 9 available on the female header (marked “9”) is a PWM capable pin. PWM allows a pin to be driven at different duty cycles. This, among many other examples, can be used to dim, or fade, an external LED. In a similar fashion as outlined in Test 1, connect an LED plus resistor to Pin 9 and ground (marked “GND”) and observe the LED fading pattern. Ensure that the LED positive terminal is in Pin 9 and that the negative terminal goes toward GND through the resistor.

6.4.5 Test 5: Analog A0 Read

The analog pin A0 is programmed to be continuously reading any input connected to it. The read input value is displayed in the Serial monitor on baud rate 9600. To see these results, load the Arduino IDE software and plug in the *Mega 2560 R3+* (ensure that the CH340 driver is installed). Turn on the Serial monitor (select baud rate 9600) and read the output window.

The user can plug a wire directly onto the female header pin A0 and the other end onto:

- A0 to Ground (marked “GND”)
- A0 to 3.3 V
- A0 to 5 V
- A0 free floating

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The output as displayed in the Serial monitor will read different values accordingly. A properly working *Mega 2560 R3+* should produce the following results:

- A0 to Ground (marked “GND”) -> Output around 0
- A0 to 3.3 V -> Output around 660, plus or minor a few
- A0 to 5 V -> Output around 1023, plus or minor a few
- A0 free floating -> Output ranges widely

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7 REFERENCES

This section lists relevant references.

- ATmega2560 datasheet by Microchip Technology:
  https://www.microchip.com/en-us/product/atmega2560

- PTSolns IDE software:
  https://PTSolns.com/IDE

- Arduino IDE software:
  https://www.arduino.cc/en/software

- PTSolns' Documentation Repository:
  https://docs.PTSolns.com

- PTSolns website:
  https://ptsolns.com/

- QWIIC connector standardization by SparkFun Electronics
  https://www.sparkfun.com/qwiic

- CH340 driver installation tutorial:

  By PTSolns:
  https://www.youtube.com/watch?v=UUQ84VKg3oM
  https://ptsolns.com/pages/ptsolns-ide-help#installing-the-ch340-driver-from-within-ptsolns-ide

  By Sparkfun:
  https://learn.sparkfun.com/tutorials/how-to-install-ch340-drivers/all

- PTSolns support:
  https://ptsolns.com/pages/contact

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